🧠 AI DECEPTION ANALYZER (CLI-BASED)

Fundamentals of AI and ML – Course Project

👨‍🎓 Submitted By

Name: Ritesh Kumar Verma
Reg No: 25BAI11426
Course: Fundamentals of AI and ML (CSA2001)

🔗 GitHub Repository:

https://github.com/codeliferitesh/ai-deception-analyzer

📌 1. TITLE

AI Deception Analyzer: A CLI-Based Machine Learning System for Detecting Deceptive Text and Emotional Tone

---

📖 2. INTRODUCTION

Artificial Intelligence (AI) and Machine Learning (ML) are becoming a major part of how systems understand human language today. From chatbots to recommendation systems, AI is everywhere.

One interesting problem is detecting whether a statement is **truthful or deceptive**. Humans often struggle with this, especially when the language is emotional or indirect.

This project, **AI Deception Analyzer**, is a simple yet effective **Command Line Interface (CLI)** application that analyzes user input text and predicts:

- Whether the statement is truthful or deceptive 🤔
- The emotional tone behind the text 😊 😡 😐

The system uses **Natural Language Processing (NLP)** techniques along with a machine learning model to perform this analysis. The main goal was to build something that is:

- Lightweight ⚡
- Easy to run in terminal 💻
- Practical and real-world focused 🌍

🚀 3. MOTIVATION OF THE PROJECT

In everyday communication, it is not easy to detect deception. People may:

- Over-explain things
- Use emotional language
- Avoid direct answers

This made me curious — *can a machine detect such patterns?*

The motivation behind this project includes:

- Exploring real-world applications of AI 🤖
- Understanding how NLP works in practice
- Building a complete CLI-based system (as required)
- Applying classroom concepts to a practical problem

❗ 4. PROBLEM STATEMENT

Detecting deception in text is a challenging task because:

- There are no clear or fixed rules
- Language is subjective and varies from person to person
- Emotional tone can hide the real intent

👉 **Problem:**
Design a system that can analyze text input and predict whether it is **truthful or deceptive**, along with identifying the **emotion behind it**.

🎯 5. OBJECTIVE OF THE PROJECT

The main objectives of this project are:

- Build a complete CLI-based AI application 💻
- Apply NLP techniques to process text 📝
- Train a machine learning model for classification 📊
- Detect emotional tone using simple logic 😊😐😡
- Store analysis history for later use 📁
- Ensure smooth and user-friendly CLI interaction

🔍 6. EXISTING METHODS

Some existing approaches for deception detection include:

1. **Human Judgment**
    - Based on intuition and experience
2. **Psychological Analysis**

    - Uses behavioral and linguistic cues
3. **Advanced AI Models (e.g., BERT)**
    - Deep learning-based approaches

⚖️ 7. PROS AND CONS OF EXISTING METHODS

✅ Pros:

- Advanced models provide high accuracy
- Psychological methods consider context

❌ Cons:

- Human judgment is often unreliable
- Deep learning models are complex and heavy
- Not suitable for simple CLI applications

💻 8. HARDWARE AND SOFTWARE REQUIREMENTS

🖥️ Hardware:

- Basic laptop or computer
- Minimum 4GB RAM

👷 Software:

- Python 3.8+
- VS Code / Terminal

📦 Libraries:

- scikit-learn
- pandas
- numpy
- joblib

⚙️ 9. METHODOLOGY AND GOAL

🛠️ Methodology:

1. Collect sample dataset
2. Preprocess text data
3. Convert text using TF-IDF
4. Train model using Logistic Regression
5. Detect emotion using keyword-based logic
6. Build CLI for user interaction

🎯 Goal:

To create a **simple, fast, and practical AI system** that works efficiently in a terminal environment.

🧩 10. FUNCTIONAL MODULES DESIGN AND ANALYSIS

- **Input Module:** Takes user input
- **Processing Module:** Converts text to features
- **Prediction Module:** Predicts deception
- **Emotion Module:** Detects emotion

- **Storage Module:** Saves results
- **CLI Module:** Handles commands

🧠 11. ALGORITHM DEVELOPMENT

1. Start program
2. Show CLI menu
3. Take user command
4. If command = analyze:
    - Take input
    - Transform text
    - Predict result
    - Detect emotion
    - Display output
    - Save history
5. If command = history:
    - Show previous results
6. If command = exit:
    - Stop program

📐 12. SOFTWARE ARCHITECTURE

User Input (CLI)
↓
Command Handler

↓
Model Processing
↓
TF-IDF Vectorizer
↓
ML Model
↓
Emotion Detection
↓
Output + Storage

👩‍💻 13. CODING

The project is implemented using Python with a modular structure:

- cli.py → CLI interaction
- model.py → Prediction logic
- train.py → Model training
- utils.py → Emotion detection

📊 14. OUTPUT

Example:

Enter command: analyze

Enter text: I swear I didn’t do anything wrong

Truth Probability : 0.30

Deception Probability : 0.70

Emotion : Neutral

🔑 15. KEY IMPLEMENTATIONS

- TF-IDF Vectorization
- Logistic Regression Model
- CLI-based interaction
- JSON-based data storage

🏆 16. SIGNIFICANT PROJECT OUTCOMES

- Built a complete AI-based CLI system
- Applied ML and NLP concepts practically
- Created a working end-to-end pipeline
- Improved understanding of real-world AI problems

🧪 17. TESTING AND REFINEMENT

- Tested with multiple inputs
- Handled invalid inputs
- Improved CLI experience
- Verified outputs

🌍 18. REAL-WORLD APPLICATIONS

- Fraud detection
- Chat/message analysis
- Customer feedback systems
- Content moderation

📌 19. CONTRIBUTION / FINDINGS

- Even simple models can give useful insights
- Language patterns are important in deception
- CLI apps are efficient and easy to use

⚠️ 20. LIMITATIONS

- Small dataset → limited accuracy
- Basic emotion detection
- Cannot fully understand human psychology

✅ 21. CONCLUSION

This project demonstrates how machine learning can be used to analyze human language and detect patterns of deception.

Although it is not perfect, it provides a strong base for future improvements and shows the practical use of AI concepts.

🔮 22. FUTURE ENHANCEMENTS

- Use advanced models like BERT 🤖

- Improve dataset size 📊
- Add voice input 🎤
- Add visual dashboard 📈
- Deploy on cloud ☁️

📚 23. REFERENCES

- scikit-learn documentation
- Python official docs
- NLP tutorials and research papers
- Online ML resources

❤️ FINAL NOTE

This project successfully fulfills all requirements:

- AI-based system ✓
- CLI-based application ✓
- Real-world problem ✓
- Structured implementation ✓

💖 ***Made with dedication by Ritesh Kumar Verma***